

Active Directory Security & Enterprise Attacks

Course Syllabus

Master Active Directory enumeration, Kerberos attacks, LDAP reconnaissance, credential attacks, privilege escalation, relay attacks, and advanced post-exploitation techniques used in enterprise penetration testing environments.

// COURSE INFORMATION

Course Information

DURATION	1 Months / 4 Weeks / 30 Hours
LEVEL	Advanced
MODULES	13
FORMAT	Hands-on Labs / Hybrid (Online + Indore Classroom)

// COURSE OVERVIEW

Course Overview

The Advanced Active Directory Penetration Testing course is designed for ethical hackers, penetration testers, red teamers, SOC analysts, and cybersecurity professionals who want to specialize in attacking and assessing Active Directory environments. This course covers LDAP enumeration, Kerberos authentication internals, Kerberoasting, AS-REP Roasting, Pass-the-Hash, Pass-the-Ticket, SMB/LLMNR relay attacks, domain enumeration, privilege escalation, credential extraction, BloodHound analysis, and complete AD attack methodology.

// LEARNING OBJECTIVES

Learning Objectives

- Enumerate Active Directory environments using LDAP and manual techniques
- Exploit Kerberos authentication (Kerberoasting, AS-REP Roasting)
- Perform Pass-the-Hash and Pass-the-Ticket attacks
- Conduct LLMNR poisoning and SMB relay attacks
- Extract credentials from SAM, SYSTEM, and NTDS.dit
- Escalate privileges from standard user to domain administrator
- Analyze attack paths using BloodHound
- Pivot through multi-subnet AD environments
- Use industry-standard tools (Mimikatz, CrackMapExec, Impacket, Evil-WinRM)

- Identify and exploit AD certificate services misconfigurations
- Provide enterprise hardening and remediation recommendations

// PREREQUISITES

Prerequisites

- Basic networking knowledge
- Familiarity with Windows environments
- Understanding of Active Directory basics
- Knowledge of Linux command line
- Basic penetration testing experience

// MODULE BREAKDOWN

Module Breakdown

01 LDAP Enumeration

- Overview of LDAP
- LDAP Enumeration Techniques
- LDAP Enumeration Tools
- LDAP Queries and Filters
- Nmap Scripts for LDAP Enumeration
- Ldapsearch Usage
- JXplorer GUI Tool
- Active Directory Structure Mapping

02 Kerberos Authentication

- Introduction to Kerberos
- Kerberos Authentication Flow
- Kerberos Encryption Types
- Ticket Granting Ticket (TGT)
- Service Tickets (ST)
- Kerberos Attack Surface
- Key Distribution Center (KDC)
- Authentication Service (AS) and Ticket Granting Service (TGS)

03 Kerberoasting Attacks

- Service Principal Names (SPNs)
- Kerberoasting Methodology
- Extracting Service Tickets
- Offline Password Cracking
- Kerberoasting Tools
- Targeted Kerberoasting
- Detection and Mitigation

04 Pass-the-Ticket (PtT)

- Kerberos Ticket Abuse
- Ticket Injection Techniques

- Lateral Movement via Tickets
- Session Hijacking
- Kerberos Ticket Manipulation
- Golden Ticket Attacks
- Silver Ticket Attacks

05 AS-REP Roasting

- Kerberos Pre-Authentication
- AS-REP Roasting Methodology
- User Enumeration for AS-REP
- Offline Password Cracking
- Identifying Vulnerable Accounts
- Kerbrute Enumeration
- Detection and Prevention

06 Manual Enumeration

- Manual Enumeration Techniques
- Operating System Enumeration
- Domain User Enumeration
- Enumerating Logged-On Users
- Enumerating Permissions and ACLs
- SPN Enumeration
- Domain Share Enumeration
- Group Membership Mapping

07 Password Attacks & Credential Dumping

- Password Attack Methodologies
- Credential Dumping Techniques
- Accessing SAM File
- Accessing SYSTEM File
- Extracting SECURITY Files
- NTDS.dit Extraction
- Offline Hash Cracking
- Password Spraying
- Hashcat and John the Ripper Usage

08 Pass-the-Hash (PtH)

- NTLM Authentication Internals
- Pass-the-Hash Attack Methodology
- Lateral Movement with PtH
- Remote Authentication Abuse
- CrackMapExec for PtH
- Mimikatz Hash Extraction
- Detection and Countermeasures

09 LLMNR Poisoning & SMB Relay Attacks

- LLMNR/NBT-NS Poisoning
- SMB Relay Attack Methodology
- Name Resolution Poisoning
- NTLM Relay Attacks
- Credential Interception
- Responder Tool Usage
- NTLMv2 Hash Capture
- Relay Attack Mitigation

10 Active Directory Offensive Security Tools

- Impacket Framework
- Ldapsearch and JXplorer
- Kerbrute
- Responder
- PowerUp
- PowerView
- BloodHound and SharpHound
- Mimikatz
- CrackMapExec
- Evil-WinRM
- Rubeus

11 Active Directory Privilege Escalation

- Windows Privilege Escalation Vectors
- Token Impersonation
- Service Exploitation
- DLL Hijacking
- Constrained Delegation Abuse
- Unconstrained Delegation Exploitation
- Resource-Based Constrained Delegation
- DCSync Attacks
- ADCS Exploitation (ESC1-ESC8)

12 Enterprise Lateral Movement & Persistence

- Lateral Movement Techniques
- Pivoting and Port Forwarding
- SSH Tunneling from Windows
- Chisel and Ligolo Usage
- Multi-Hop Pivoting
- Golden and Silver Ticket Forging
- Shadow Credentials
- Persistence Mechanisms
- Domain Trust Exploitation

13 Active Directory Hardening & Reporting

- AD Security Best Practices
- Kerberos Hardening
- NTLM Restriction Policies
- Privileged Access Management
- Detection Strategies
- SIEM Integration for AD Attacks
- Penetration Test Reporting
- Remediation Recommendations
- Enterprise Security Posture Assessment

// TOOLS & HANDS-ON LABS

Tools & Hands-On Labs

- Multi-domain Active Directory forest
- Multiple organizational units with various permissions
- Certificate Services (ADCS) lab
- Multi-subnet network with segmentation
- Realistic enterprise user and group structure
- Kerberos attack simulation targets
- Relay attack practice environment
- BloodHound data collection and analysis lab

// TRAINING MODE

Training Mode

Every Armour Infosec course runs as a unified programme delivered in two parallel modes — the same curriculum, the same trainers, the same certification, regardless of how you join.

- ✓ Online Live Classes — real-time, instructor-led, fully interactive sessions
- ✓ On-Premise Classroom Training — in-person at our Indore centre (Sudama Nagar)
- ✓ Both modes run concurrently in every batch; switch between them as your schedule needs
- ✓ Same syllabus, lab access, and certification track for online and on-premise students

// CERTIFICATIONS & CAREER OUTCOMES

Certifications & Career Outcomes

This course aligns with industry-recognised certifications and prepares graduates for offensive-security, application-security, and infrastructure-security roles.

- OSCP+ AD module — Advanced Offensive Security Certification
- CEH (Certified Ethical Hacker)
- CRTP (Certified Red Team Professional)
- CRTE (Certified Red Team Expert)
- HTB Pro Labs (Offshore, RastaLabs)
- PNPT (Practical Network Penetration Tester)

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Reach out to discuss enrolment, batch schedule, and lab access. Our Indore training centre runs both in-person and live online cohorts with placement assistance.

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